TriZOD — Final Pipeline & Re-Referenced Dataset

Finalized pipeline · α-synuclein case study · chemical-shift perturbations

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TUM · TriZOD project meeting

TriZOD pipeline at a glance

BMRB
NMR-STAR archive
17,388 entries

→

Parser
bmrb.py
methyl wildcards
Leu CDx · Val CGx

cache: tmp/bmrb_entries/*.pkl

→

Filter
4 stringency tiers
keyword + paramag
+ shift-coverage

entries passing
- unfilt 16,851
- tolerant 15,433
- moderate 10,107
- strict 3,033

→

Re-referencing
LACS (CA, CB, C′, HA, H, N)
then **POTENCI / AIC**
residual (HB + others)

--rereference-mode
{none, lacs, potenci-only, **both**}

cache: tmp/potenci/*.json · tmp/wSCS/*.npz

→

Scoring
backbone-only
Z-score (signed)
G-score (3-residue triplet, geom. mean)

→

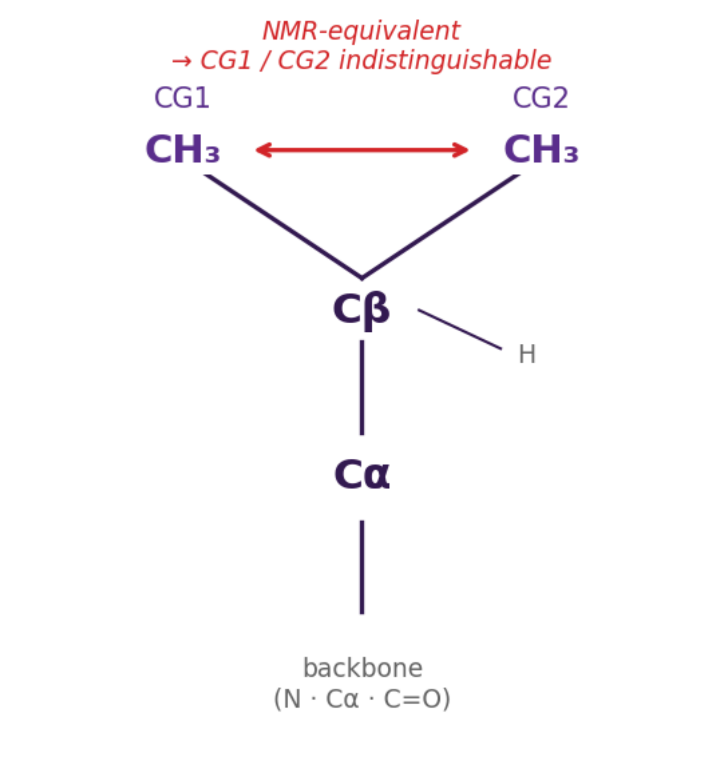
Output
scores.json
*_rereferenced.str
Zenodo metadata

--emit-str <dir>

Next steps:
- exclude multi-molecule (bound) entries that distort G-scores
- per-sequence representative pick (best conditions → median G-score)
- mmseqs2 sequence clustering for ML train/val/test split

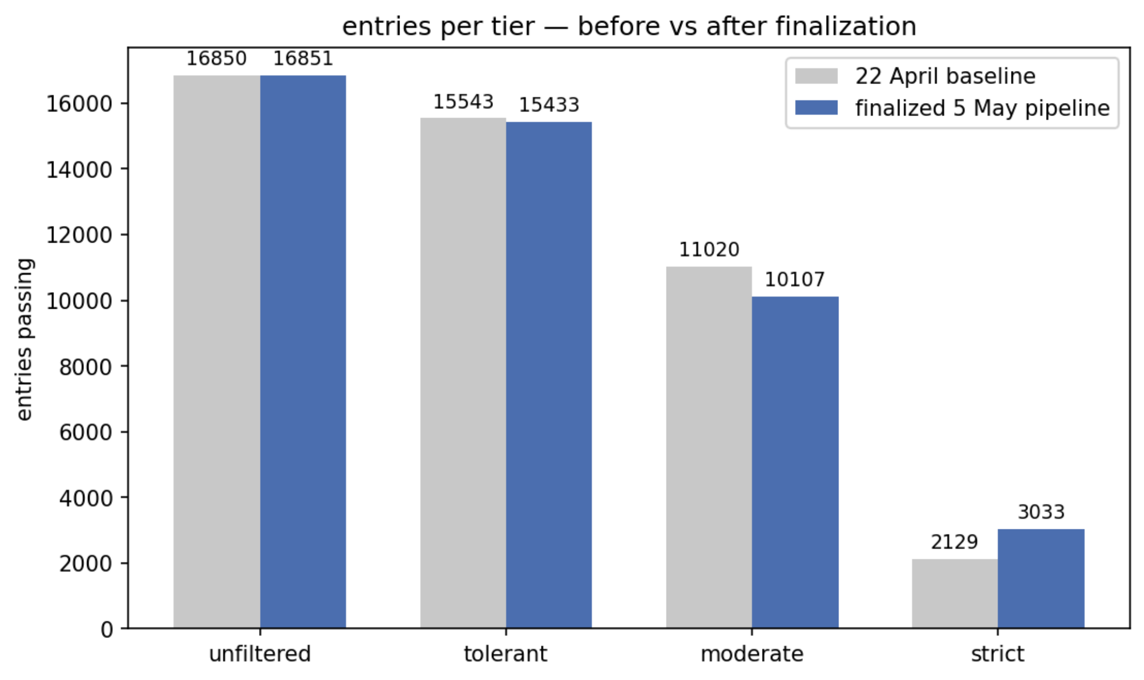
Implementation update

- Methyl wildcards in the parser (Leu `CD1/CD2` → `CDx`, Val `CG1/CG2` → `CGx`)
- LACS pre-correction baked into the scoring pipeline
- New flag: `--rereference-mode {none, lacs, potenci-only, both}` (default `both`)
- New flag: `--emit-str <dir>` writes one re-referenced NMR-STAR per scored entity



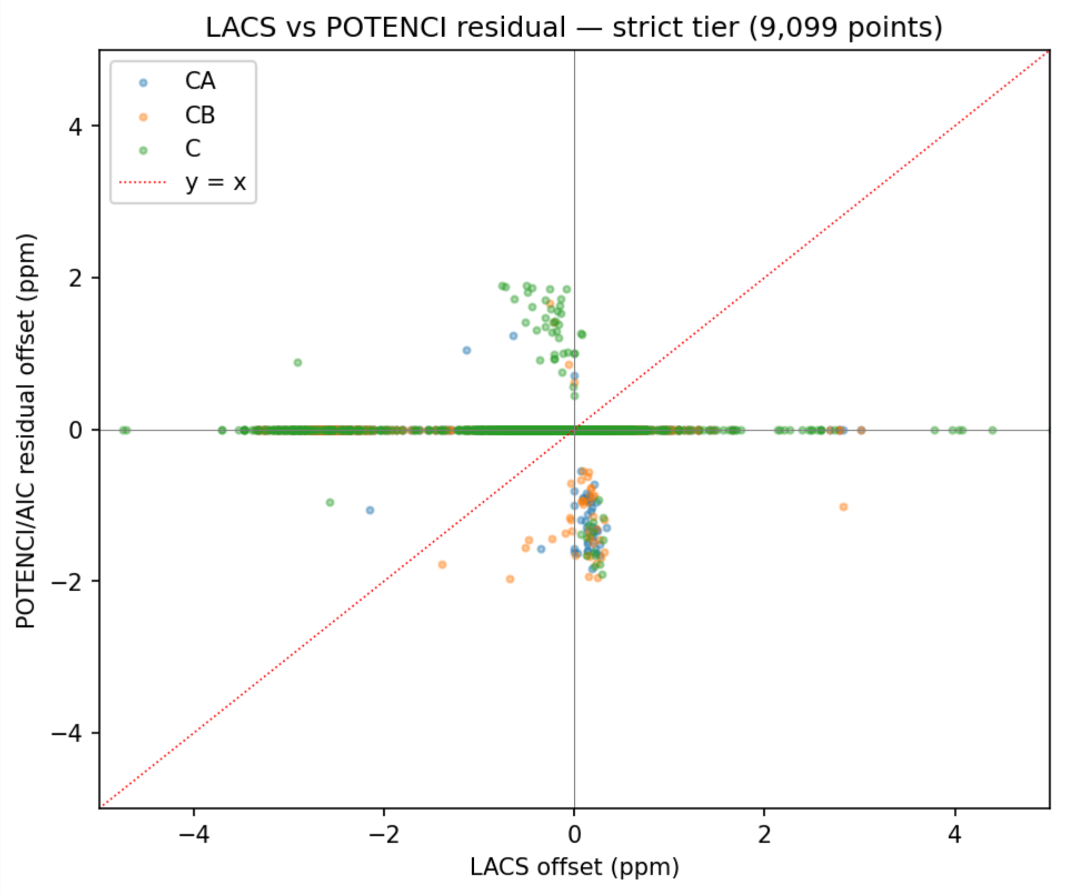
Valine side chain — geminal methyls

Filter improvements and per-tier dataset deltas



Removed denaturant false-positives, fixed the solid-state regex, added a paramagnetic-sample filter, relaxed `min-backbone-shift-types` 5→4 in strict, and broadened the Celsius heuristic.

LACS vs POTENCI residual capture

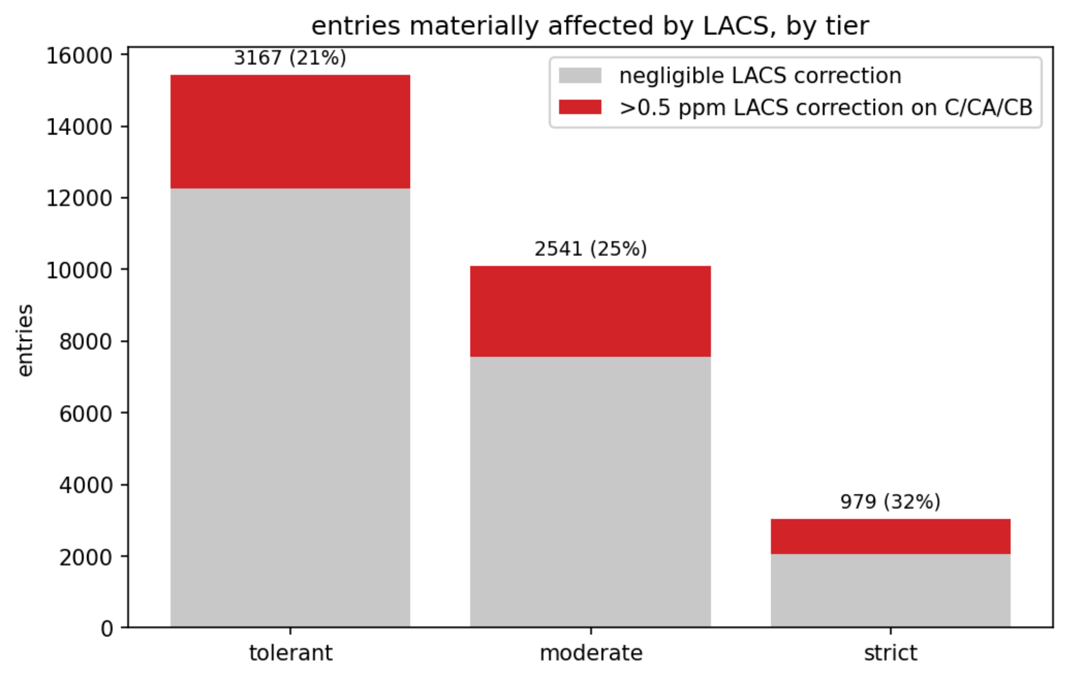


Each dot is **one entry × one atom** (CA / CB / C, strict tier).

LACS catches the large systematic referencing bias.

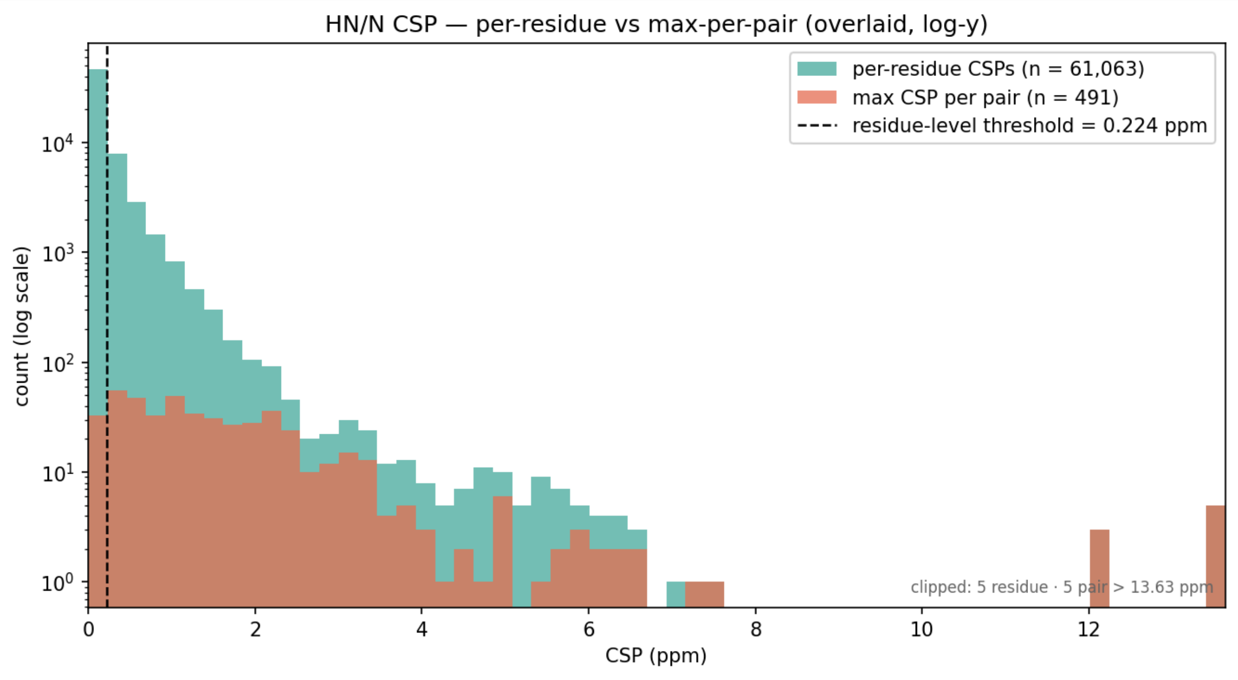
POTENCI/AIC residual mops up the remainder — most points land near the origin once LACS has done its job.

Entries affected by LACS



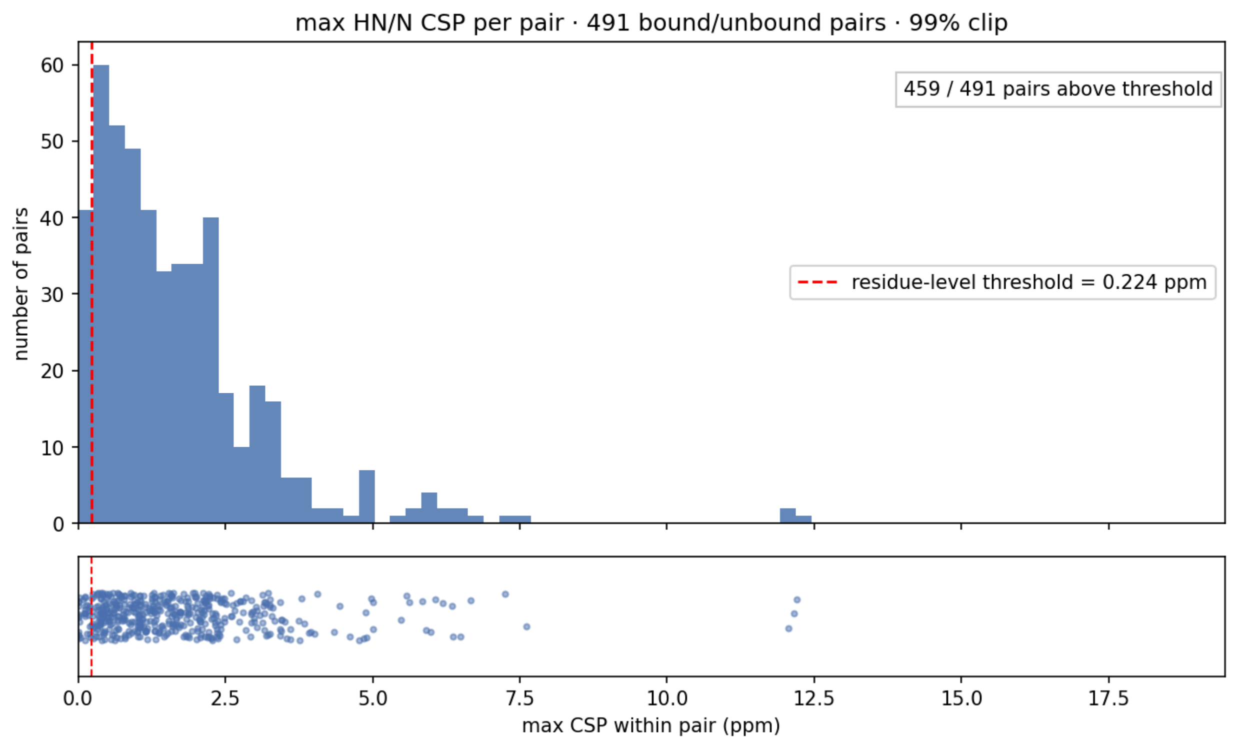
Red = entries with $|\mathrm{LACS\ offset}| > 0.5$ ppm on at least one of C/CA/CB. Re-referencing meaningfully changes the input shifts for **21%** (tolerant), **25%** (moderate), **32%** (strict).

Chemical-shift perturbation distribution



Teal: 61,063 per-residue HN/N CSPs · **orange:** one max CSP per pair (n = 491). Threshold **0.224 ppm** = trimmed-mean+SD on the bottom 90%. Most residues sit below it; almost every pair has at least one residue above. 90 of 581 candidate pairs dropped — no overlapping HN+N coverage between the two entries.

Max CSP per pair · how strong is the strongest binding shift?



One value per pair (491 points): the **maximum** HN/N CSP within each pair. Pairs below the 0.224-ppm threshold are essentially silent; those above are real binding events.

What's still TODO

- **Multi-molecule entries distort G-scores** — exclude entries with a non-polymer or nucleic-acid binding partner from the dataset.
- **Per-sequence representative selection** — for the same protein with multiple BMRB entries, pick the one with best experimental conditions; tiebreak by median G-score.
- **mmseqs2 sequence clustering** for the ML train/val/test split (per the original CheZOD/TriZOD report).
- All three are flagged as TODO in the workflow on slide 2.

Released artefacts

- `data/release/<tier>/scores.json` — per-residue Z/G + LACS + POTENCI offsets
- `data/release/<tier>/str/bmr*_*_rereferenced.str` — re-referenced NMR-STAR
- `.zenodo.json` + `CITATION.cff` — DOI on first tagged release

Per-tier counts after the rerun:

Tier	Entries	Coverage
strict	3,033	17.00%
moderate	10,107	56.64%
tolerant	15,433	86.49%
unfiltered	16,851	94.45%

Next steps

- Push deposit to Zenodo (DOI placeholder until first tag)
- Iterate on Reid's CSP — extend to multi-atom $\Delta\omega_{\mathrm{RMS}}$ (Schumann/ Williamson 2008)
- Reid follow-up with John Markley on why BMRB stopped LACS reports (frozen July 2020)
- Possible auto-assignment partner who'd benefit from `CDx`/`CGx` outputs